



Bahir Dar Institute of Technology
Bahir Dar University
Faculty of Computing
Departments of Software Engineering
Final year Project Titles

- **AI-Powered SaaS PPT Slide Generator**
- **AI-Powered Full-Stack Website Builder**
- **UI/UX Mock – Full-Stack AI Design SaaS Platform**

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1. AI-Powered SaaS PPT Slide Generator

1.1 Introduction

Presentations are essential in academic, corporate, and entrepreneurial contexts. Creating professional slides is time-consuming and requires structured thinking, design skills, and visual asset integration. Users often spend more time formatting slides than refining their message. Existing tools rely on manual editing or unstructured AI outputs, which still require substantial effort. A more intelligent, automated solution is needed.

1.2 Business Problem

Inefficiency in presentation creation affects productivity:

- Manual formatting and layout adjustments
- Lack of automated slide structure generation
- Poor integration between AI content and design
- Time-consuming editing processes
- No seamless workflow from idea to export

Students, professionals, and startups under tight deadlines face delays and inconsistent presentation quality.

1.3 Proposed Solution

The **AI-Powered SaaS PPT Slide Generator** converts a simple user prompt into a structured, professionally designed PowerPoint presentation. The platform:

- Interprets prompts with Generative AI
- Generates structured slide outlines and bullet content
- Applies consistent design themes
- Integrates visuals automatically
- Provides real-time editing
- Exports fully editable .pptx files

This approach automates both content and design while allowing human review.

1.4 System Architecture

1.4.1 Frontend

- React + Vite ensures responsive performance
- Tailwind CSS maintains consistent styling
- Modular components support live slide editing

1.4.2 Backend & AI

- Node js for backend development
- Firebase Firestore handles secure data storage
- Gemini AI via Firebase Genkit generates structured slide outlines

1.5 Target Users

- Students preparing academic presentations
- Corporate professionals creating reports and decks
- Startup founders developing investor pitches
- Marketing and consulting teams

1.6 Conclusion

By automating content structuring and visual formatting, this platform reduces preparation time while ensuring professional output. Real-time editing and export capabilities make it a practical, scalable solution for modern presentation needs.

2. AI-Powered Full-Stack Website Builder

2.1 Project Overview

A SaaS platform enabling users to generate fully responsive, production-ready websites using simple natural language prompts. Built on the **PERN stack (PostgreSQL, Express, React, Node.js)** the platform ensures scalability and maintainability.

An LLM via OpenRouter API interprets prompts and generates HTML/Tailwind CSS instantly. Users refine sites via AI conversational updates or a visual Editor Panel (text, colors, layout, typography). **Neon PostgreSQL** ensures data reliability and serverless scaling.

2.2 Business Problem

Traditional web development is complex and resource-intensive:

- **High Technical Barrier:** Most users lack coding skills
- **High Cost and Time:** Developer or agency services are expensive
- **Iteration Difficulty:** Minor updates require technical support
- **Fragmented Tools:** Separate platforms for design, hosting, and authentication

A unified AI-driven platform is needed to simplify website creation for non-technical users.

2.3 Objectives

- Instant website generation from text prompts
- Hybrid editing: AI-driven and manual adjustments
- Community engagement via public project gallery
- Scalable performance supporting concurrent users

2.4 Technical Architecture

2.4.1 Frontend

- React , Tailwind CSS
- React Router DOM for navigation
- IFrame-based previews (desktop, tablet, mobile)

2.4.2 Backend

- Node.js + Express REST API
- Better Auth for secure authentication
- Prisma ORM + Neon PostgreSQL for database management

2.4.3 AI & Integrations

- OpenRouter API for AI-generated code

- Axios for frontend-backend communication

2.5 Key Features & Workflow

- AI-enhanced prompt website generation
- Project dashboard for view, edit, restore, delete
- Hybrid editing with AI or manual interface
- Real-time responsive previews
- Export source code or publish to community gallery

2.6 Target Users

- Students building academic websites
- Entrepreneurs and startups creating landing pages or portfolios
- Corporate teams managing internal or client web content
- Marketing teams needing rapid site production

2.7 Conclusion

The AI-Powered Website Builder removes technical barriers while preserving creative control. AI-driven generation, hybrid editing, and integrated monetization offer a scalable, collaborative, and professional platform for modern web development.

3. UI/UX Mock – Full-Stack AI Design SaaS Platform

3.1 Project Overview

UI/UX Mock generates professional UI/UX designs for web and mobile using simple prompts. It outputs production-ready HTML/Tailwind code, supports multi-screen layouts, and offers an interactive drag-and-drop canvas. Users can switch themes dynamically and export designs as PNGs or raw code.

3.2 Business Problem

Design is a major bottleneck:

- High cost of professional services
- Time-consuming manual prototyping
- Technical gap between visual design and functional code
- Limited scalability for multiple iterations

Non-technical users face delays and high costs.

3.3 Project Objective

Automate the design-to-code workflow:

- Generate high-fidelity HTML/Tailwind designs from prompts
- Apply theme-based customization (Netflix, Spotify, Amazon)
- Enable interactive canvas editing
- Provide seamless export (PNG, raw code)
- SaaS model with authentication and subscription tiers (Free/Unlimited)

3.4 Technical Architecture

3.4.1 Frontend

- Tailwind CSS + Shadcn UI
- Interactive canvas for layout management

3.4.2 Backend

- Neon PostgreSQL + Drizzle ORM for serverless, scalable data management
- Node.js framework

3.4.3 AI Integration

- OpenRouter API for prompt interpretation and design generation

3.5 Key Features & Workflow

- Prompt-to-design generation
- Interactive canvas manipulation
- Dynamic theme switching
- Export as PNG or raw code

- Project dashboard for management

3.6 Target Users

- Developers needing quick prototypes
- Startups launching MVPs
- UX/UI designers generating concepts
- Marketing teams producing multiple design iterations

3.7 Conclusion

UI/UX Mock transforms design from a manual, time-intensive process into an AI-powered, fully automated workflow. It democratizes high-end design, accelerates product development, and enables anyone to build professional digital products quickly.